Technische Universität München

# Developing an Inventory Management System for Boeing R&D

## A Centralized, Traceable Solution for Drilling Test Operations

David Vogenauer, Divya Solanki, Siddhesh Chavan, Simon Karan, Sonja Lind, Sude Karataş

### Project Context: R&D Drilling Operations

- A Boeing airplane has >2 million drilled holes
- Boeing conducts extensive R&D on drilling metal and composite materials
- This is done in collaboration with TUM Smart Drilling Cell (Garching)

### The Problem: Outdated Asset Tracking

Inefficient inventory management of specimens, cutting tools, and measurement instruments.

- Current solution: Excel files
- Lack of shared, centralized management system, leading to data loss
- Assets do not have standardized IDs and latest stock is unknown
- Reservation and tracking is a challenge

**1** Receive shipment

**2** Enter asset into system

Barcode for identification

**3** Stored in centralized database

Automatic Notifications
- End of asset life
- Calibration date
- Replenish stock

**4** Desktop System
- Search storage inventory
- Reserve assets for projects
- Trace asset location

**5** Mobile System

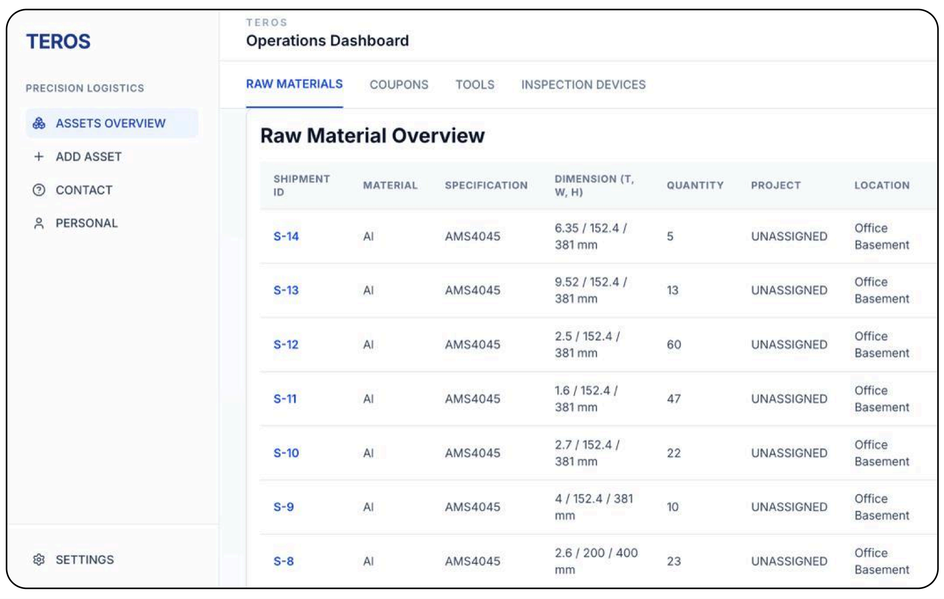
Scan barcodes to access details

Teros by TUM x Boeing

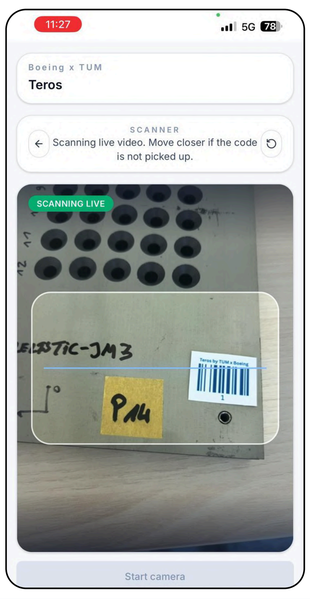
### Technologies Used

React, JavaScript, Next.js, Python, Flask, SQLite, Vercel, aws

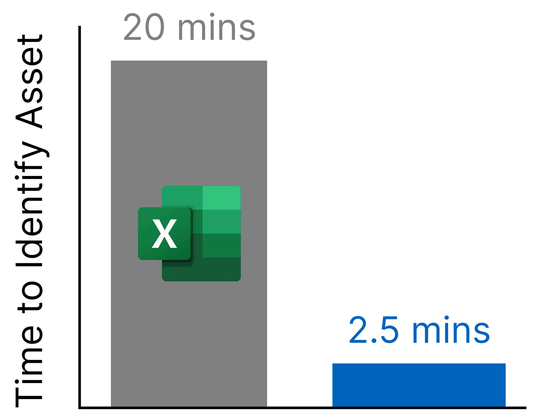
### Desktop Dashboard




### Barcode Scanner




### Impact




**€14.000**
Saved per Year

**100+**
Assets Onboarded

**8-10**
Active Users at Boeing & TUM

### Deliverables

1. Database Architecture
2. Desktop and Mobile App
3. KPI Benchmark
4. User Manual

**Try Scanning an Asset!**

**Technische Universität München**
School of Computation, Information and Technology (CIT)
1.000+ Project Week

BOEING
Technische Universität München